

CEP mini project – Group No. 9

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Topic: Air Quality Index Analysis & Heat Prediction.

SDG 13 – Climate Action.

Problem:-

Expanding concrete infrastructure and heavy construction have created heat trapping cities stripped of tree cover, while vehicle emissions, industrial output, construction dust, and open waste burning continually degrade local air quality. This thermal stress and pollution are compounded by sparse AQI sensors, restricted environmental data, language barriers, and a total lack of heat-aware navigation for commuters. Furthermore, citizens receive conflicting guidance and lack direct reporting channels, allowing unchecked environmental degradation to damage soil, harm biodiversity, accelerate infrastructure wear, and ultimately drain nearly 2% of national GDP annually in lost labour productivity. *(Finalized on 09.08.2026)*

Problem statement:-

There is a critical need for an accessible, accurate, and unified system that monitors real-time AQI and heat data, maps cooler and safer routes, raises public awareness on effective heat-mitigation practices, and bridges the information gap between government data sources and citizens — in order to reduce health risks, environmental degradation, infrastructure damage, and economic losses caused by rising urban heat and poor air quality.

Problems associated with our topic :-

Problem 1 – Related to heat.

- a) We think that the city is becoming smarter and developing by building high concrete roads but it is not like that.
- b) These concretes (Roads) **trap lots and lots of heat** into them in the peak sun hours and also in the morning, which makes the atmosphere at the night hot and warm as the heat trapped/absorbed at the day time is released at the night time.

Problem 2 – Decreasing number of trees/bushes on routes.

Travellers, citizens, tourists, etc. **feel heat exhaustion** due to less greenery, concrete roads and decreasing number of shading trees on route.

Problem 3 – Failing sensor technology.

- a) Sensors monitoring the AQI of a particular city are **placed miles apart** and are less in count.
- b) Sensors show **wrong/ Less appropriate AQI readings** which causes confusing in analysing the air quality of a particular region.

Problem 4 – Language Barrier & data accessibility.

Normal people, citizens don't have access to world wide data about AQI and heat parameters. Government sites don't show the whole data to the user other than government servants.

Problem 5 – Lack of practical awareness.

People don't know if they have a heat problem or any Air related issues where to contact and where to inform or report.

Problem 6 – No knowledge about heat mapping.

- a) When ever people travel in summers, they are exhausted due to the traffics and on top of that the heat poured by the sun.
- b) Also, they cannot locate routes that are cooler, accessible and less warm.

Problem 7 – Damage soil quality, reduces photosynthesis and harms biodiversity.

Basically, damages mother earth.

Problem 8 – Constructional damage due to heat.

Prolong heat softens asphalt expands rails and bridges and accelerates age of building materials, increases maintenance cost.

Problem 9 – Citizens getting mixed suggestions/solutions.

- a) Some of the NGOs, tell the people that close your windows to save yourself from the heat traps.
- b) Some say keep your windows open for ventilation and coolness.
- c) This creates confusion between the people whether to do what?

Problem 10 – Economic loss at national scale.

- a) Extreme heat is estimated to cost India the equivalent of 2% of GDP in lost wages and productivity every year.
- b) Lost working hours run into billions of potential labour hours annually.

(Finalized on 06.08.2026)
